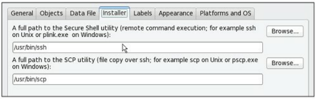

Figure 12: Setting up the environment

the active node's internal cache should display the same number of entries as the backup's external cache, and vice-versa. To verify that the recovery works fine, when you trigger a fail-over, the log files should display the following information:

```
[root@Firewall1 ~]# conntrackd -s
cache internal:
current active connections:          3
connections created:                23    failed:         0
connections updated:                24    failed:         0
connections destroyed:              20    failed:         0

cache external:
current active connections:          3
connections created:                22    failed:         0
connections updated:                13    failed:         0
connections destroyed:              19    failed:         0

traffic processed:
              985767 Bytes                3018 Pckts

multicast traffic (active device=eth1):
              194424 Bytes sent         193032 Bytes recv
               12038 Pckts sent          11989 Pckts recv
                 106 Error send              0 Error recv

message sequence tracking:
                   0 Msgs mfrm             133 Msgs lost
```

Run *conntrack –E*, which displays the connection tracking events:

```
[root@Firewall1 ~]# conntrack -E
    [NEW] tcp      6 120 SYN_SENT src=192.168.170.1 dst=10.1.1.10 sport=1123
dport=22 [UNREPLIED] src=10.1.1.10 dst=192.168.170.1 sport=22 dport=1123
 [UPDATE] tcp      6 60 SYN_RECV src=192.168.170.1 dst=10.1.1.10 sport=1123
dport=22 src=10.1.1.10 dst=192.168.170.1 sport=22 dport=1123
 [UPDATE] tcp      6 432000 ESTABLISHED src=192.168.170.1 dst=10.1.1.10
sport=1123 dport=22 src=10.1.1.10 dst=192.168.170.1 sport=22 dport=1123
[ASSURED]
```

Or, run the code shown below:

```
[root@Firewall1 ~]# conntrack -L -p tcp --dport 22
tcp      6 425441 ESTABLISHED src=192.168.170.1 dst=192.168.170.252
```

Figure 13: Pushing the policy

```
sport=1541 dport=22 packets=1705 bytes=89624 src=192.168.170.252
dst=192.168.170.1 sport=22 dport=1541 packets=1771 bytes=487161 [ASSURED]
mark=0 secmark=0 use=2
tcp      6 299 ESTABLISHED src=192.168.170.1 dst=192.168.170.252 sport=1080
dport=22 packets=600 bytes=43404 src=192.168.170.252 dst=192.168.170.1
sport=22 dport=1080 packets=521 bytes=90397 [ASSURED] mark=0 secmark=0
use=2
tcp      6 431696 ESTABLISHED src=192.168.170.1 dst=10.1.1.10 sport=1123
dport=22 packets=66 bytes=3004 src=10.1.1.10 dst=192.168.170.1 sport=22
dport=1123 packets=62 bytes=6696 [ASSURED] mark=0 secmark=0 use=2
```

You can dump the internal cache with the following command:

```
[root@Firewall1 ~]# conntrackd -i
tcp      6 ESTABLISHED src=192.168.170.1 dst=192.168.170.252 sport=1080
dport=22 src=192.168.170.252 dst=192.168.170.1 sport=22 dport=1080
[ASSURED] [active since 3258s]
tcp      6 ESTABLISHED src=192.168.170.1 dst=192.168.170.252 sport=1541
dport=22 src=192.168.170.252 dst=192.168.170.1 sport=22 dport=1541
[ASSURED] mark=0 secmark=0 [active since 1274664626s]
tcp      6 ESTABLISHED src=192.168.170.1 dst=10.1.1.10 sport=1123 dport=22
src=10.1.1.10 dst=192.168.170.1 sport=22 dport=1123 [ASSURED] [active since
498s]
```

END

References

- http://conntrack-tools.netfilter.org
- http://www.keepalived.org
- http://www.fwbuilder.org

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